

# Tom Cantellow

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## EXPERIENCE

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### Senior Scientific Officer - R&D Fellow

Aug. 2025 – Present

Department for Environment, Food & Rural Affairs

Newcastle, UK

- Built ETL pipelines and reproducible analytical workflows using R, Python, and SQL, processing 8,000+ survey responses through Databricks and Azure infrastructure with delta table management and automated validation.
- Translated complex technical analysis for diverse stakeholders including policy officials and senior leadership, presenting actionable insights and managing project delivery across competing priorities.
- Convened cross-disciplinary steering group spanning policy, data science, and behavioural insights, shaping project strategy and establishing data governance standards and best practices.

### Research Associate

Apr. 2023 – Dec. 2024

University of Bristol

Bristol, UK

- Developed Net-Zero Resilience Index using highly technical eigenvector-based indices, proximity matrices, and large-scale employment datasets alongside machine learning to analyse employment records, producing interactive dashboards across multiple geographies. Identified regional economic vulnerability patterns and generated policy-relevant insights.
- Presented findings to 8+ government departments, translating technical analysis into accessible reports and policy briefings, resulting in ministerial briefings and parliamentary evidence submission. Peer-reviewed journal article published.
- Built data integration workflows merging Business Register Employment Survey, Standard Industrial Classification codes, and labour market indicators, ensuring data quality and interoperability.

### Teacher

Oct. 2020 – May 2025

University of Bristol

Bristol, UK

- Designed and delivered quantitative analysis lectures teaching Excel, R, and data visualisation to diverse undergraduate cohorts, adapting communication for technical and non-technical audiences.
- Mentored students on analytical approaches, dataset construction, and statistical methods, building analytical capability and promoting best practices in data handling and presentation.

## EDUCATION

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### University of Bristol

Bristol, UK

PhD in Human Geography (Quantitative Methods)

Oct. 2020 – Nov. 2024

### University of Sheffield

Sheffield, UK

MSc, Distinction, in Applied GIS

Sep. 2017 – Sep. 2018

### University of Lincoln

Lincoln, UK

BA (Hons), 2:1, in International Relations

Sep. 2012 - May 2016

## KEY PROJECTS

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### Net Zero Resilience Index | R, Python, Machine Learning, BI Dashboards

Jun. 2024 – Dec. 2024

- End-to-end analytical pipeline with data ingestion, cleansing, statistical analysis, and interactive visualisation using R Shiny and Leaflet; peer-reviewed journal article published.
- Applied machine learning classification, PCA, and spatial econometrics across 365 Local Authorities; produced publication-quality visualisations and user-centric interactive tools tailored to policy audiences.

### PhD Thesis: Voting, Inequality & Political Values | R, SQL, Statistical Modelling

Sep. 2020 – Nov. 2024

- Authored a 58,000-word peer-reviewed PhD thesis analysing UK voting behaviour and how the operationalisation of neoliberalism shapes inequalities, political values, and personal financial wellbeing.
- Integrated large datasets (British Election Study, Land Registry, Census) using mixed-effects regression and ETL workflows with data quality assurance and validation.

## TECHNICAL SKILLS

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**Languages & Tools:** R, Python, SQL, Excel, Power BI, Tableau, RStudio, Databricks, Azure, QGIS, ArcGIS, Git

**Expertise:** ETL pipelines, End to End research design, statistical methods, digital mapping, spatial analysis, machine learning, data quality assurance, dashboards, stakeholder management, RAP principles.